

MyTix System Limitations and Improvement Opportunities

Current-state assessment and practical roadmap

Scope

This document assesses the implemented Java/MySQL console application. It separates present capabilities from gaps visible in the user interface and source implementation, so improvements can be planned without overstating current functionality.

Current strengths

- Clear role separation for guests, customers, and organizers.
- Relational constraints and transactions support account creation, booking, resale, and performance cancellation.
- The system supports reserved and general-admission sales, resale price caps, reviews, blocking, refunds, searches, and reports.
- Organizer ownership checks are applied to event and performance management operations.

Key limitations

Area	Current limitation	Impact
Authentication	Login matches an ID or email and account type; no password, session control, or recovery flow is present.	Anyone knowing an identifier may be able to access an account.
Payments	Card data is collected and stored directly; no payment gateway, tokenization, encryption evidence, or payment status workflow is exposed.	High security and compliance risk; payments are not production-ready.
Interface	The application is command-line only and relies on exact input formats, IDs, names, and dates.	Harder for nontechnical users; invalid input can interrupt flows.
Discovery	Several lookups require exact titles, names, or addresses; results are printed rather than navigable.	Users may struggle to find the intended record.
Operations	No visible notifications, ticket delivery/QR codes, check-in, waitlist, or customer support workflow.	Incomplete end-to-end event attendance experience.
Data quality	Some management tasks are single-item operations, such as assigning one section/tier at a time.	Setup can be slow and mistakes are harder to detect.
Reporting	Reports print raw rows to the console without export, charts, scheduled delivery, or access controls.	Limited usefulness for business decision-making.
Recommendations	Pricing suggestions use simple historical averages and limited comparable criteria.	Recommendations can be inaccurate or biased when demand differs.

Recommended improvements

Priority	Improvement	Expected benefit
1 - Security	Implement password hashing, account verification, role-based authorization, secure sessions, audit logging, and least-privilege database credentials. Replace stored card details with a PCI-compliant payment provider and tokens.	Protects users and makes the platform suitable for real accounts and payments.
2 - Reliability	Add centralized input validation, user-friendly error handling, idempotent booking/resale operations, transaction isolation/concurrency controls, and automated backup/recovery procedures.	Reduces crashes, duplicate sales, and operational loss.
3 - Usability	Build a web or mobile interface with searchable listings, form validation, seat maps, account pages, and clear confirmations. Support partial matching and filters that preserve user choices.	Makes core workflows accessible and discoverable.
4 - Ticket lifecycle	Issue digital tickets with QR/barcodes, add venue check-in/scanning, email confirmations, alerts for cancellations/refunds/resale activity, and configurable cancellation policies.	Completes the customer journey and improves event operations.
5 - Organizer tools	Provide multi-section configuration, templates, preview/validation before publishing, inventory dashboards, and exports.	Speeds event creation and lowers setup errors.
6 - Analytics	Add dashboards, CSV/PDF export, report scheduling, data retention rules, and explainable demand/pricing models based on broader comparables.	Turns stored data into actionable insights.

Suggested delivery sequence

- First: secure authentication and payment handling; add automated tests for booking, cancellations, and resale.
- Next: improve concurrency protection and exception handling, then provide a user-facing web interface.
- Then: add ticket delivery/check-in and notifications.
- Finally: enhance reporting, forecasting, and organizer optimization after trustworthy operational data has accumulated.

Success measures

Track successful booking completion, failed/duplicate transaction rate, support requests per booking, time to configure an event, refund completion time, check-in rate, and report adoption. Security work should also include periodic access reviews, dependency updates, and tested recovery exercises.

Conclusion

MyTix has a solid academic prototype foundation: its relational workflows cover the principal ticketing concepts. The highest-value next steps are safeguarding identities and payments, protecting transactions under load, and replacing the console-only experience with a guided interface.